# Exact all-parameter spectral-gap certificates for deformed-EPR Hamiltonians on small subdivided stars

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**Abstract**

We study the coefficient-one deformed-EPR Hamiltonian $H_s = -\sum_e (|00\rangle + s|11\rangle)(\langle 00| + s\langle 11|)_e$ on finite trees, where $0 < s < 1$ and the rank-one edge term is not normalized. A recent conjecture gives the lower bound $\Delta(H_s) \geq 1 - s^2$, with equality on stars. We give an exact finite-family result for the subdivided star $T_q$, obtained from a star by subdividing one edge: for every $q = 1, \ldots, 6$ and every real $0 < s < 1$,

$$\Delta(T_q, s) > 1 - s^2.$$

The case $q = 1$, the four-vertex path, is proved by an explicit quartic root-ordering argument. For $q = 2, \ldots, 6$, permutation symmetry and a conserved staggered occupation reduce the full Hamiltonian to real symmetric blocks of dimension at most eight. Exact characteristic polynomials and crossing resultants over $\mathbb{Q}[t, x]$, with $t = s^2$, reduce a possible failure to 171 algebraic parameter neighborhoods. Rational root isolation and Weyl-continuity enclosures prove a positive margin on every neighborhood, while an exact reference sign fixes each remaining component. The result excludes these six non-star trees as additional equality cases. It does not prove the conjecture for arbitrary trees or for $q \geq 7$.

## 1 Introduction

Wong, Bravyi, Gosset and Liu introduced a family of two-local Hamiltonians whose edge interactions favor a deformed EPR vector and connected its spectral properties to Lee–Yang tensors [1]. In the coefficient-one convention used here, the Hamiltonian on a finite graph $G = (V, E)$ is

$$H_s(G) = -\sum_{\{i,j\} \in E} \big(|00\rangle + s|11\rangle\big)\big(\langle 00| + s\langle 11|\big)_{ij}, \qquad 0 \leq s < 1. \tag{1}$$

This model lies in the Suzuki–Fisher class [2]. The source proved uniqueness and even parity of the ground state and found the exact gap $1 - s^2$ on every star with at least three vertices. It then conjectured the lower bound

$$\Delta(G, s) \geq 1 - s^2 \tag{2}$$

for every finite connected graph. Its evidence included exact diagonalization at 100 uniformly spaced parameter values for all trees with at most 18 vertices and all connected graphs with at most 10 vertices, as well as a mixed exact-diagonalization and matrix-product-state study of paths [1]. Those finite grids are strong evidence, but do not order the spectrum over an uncountable parameter interval.

Two subsequent Lee–Yang gap results give broad but differently normalized bounds. Bravyi, Gosset, Liu and Wong proved a gap of at least twice the minimum longitudinal field for the

Suzuki–Fisher family [3]. Using the Pauli expansion of (1), this gives $\Delta \geq \frac{1}{2}(1-s^2)\delta(G)$, where $\delta(G)$ is the minimum degree. It therefore supplies only half of (2) on a tree. Rayudu and Takahashi proved another uniform-field Lee–Yang gap bound [4]. Neither result provides the sharp all-parameter ordering established below for the specified finite family.

Our graph $T_q$ has ordinary leaves $L_1, \ldots, L_q$, a centre $C$, and two further vertices $A, B$, with edges

$$CL_i \quad (1 \leq i \leq q), \qquad CA, \qquad AB. \tag{3}$$

Thus $T_q$ has $q+3$ vertices and $q+2$ edges; it is obtained by subdividing one edge of the star $K_{1,q+1}$. Every $T_q$ is a non-star, including $T_1 = P_4$. The main result is deliberately finite.

**Theorem 1** (Finite subdivided-star result)**.** *For each integer $q \in \{1,2,3,4,5,6\}$ and every real $0 < s < 1$, the Hamiltonian (1) on $T_q$ satisfies*

$$\Delta(T_q, s) > 1 - s^2. \tag{4}$$

The restriction in $q$ is part of the theorem, not a finite-size proxy for an asymptotic claim. The proof is analytic for $q = 1$ and exact computer-assisted for $2 \leq q \leq 6$. It controls both the lowest odd state and the second even state. This is essential: controlling only the odd excitation would assume a separate conjecture made in Ref. [1].

## 2 Hamiltonian, normalization and gap channels

Put $t = s^2$ and work with the positive semidefinite operator

$$K_s = -H_s = \sum_{e \in E} |\phi_s\rangle\langle\phi_s|_e, \qquad |\phi_s\rangle = |00\rangle + s|11\rangle. \tag{5}$$

In the ordered basis $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$, one edge term is

$$|\phi_s\rangle\langle\phi_s| = \begin{pmatrix} 1 & 0 & 0 & s \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ s & 0 & 0 & s^2 \end{pmatrix}. \tag{6}$$

It has rank one and sole nonzero eigenvalue $1 + s^2$. In particular, it is not a projector in the convention of (1). Energies and gaps below are dimensionless, with coefficient one on every edge.

The parity $P = Z^{\otimes |V|}$ commutes with $K_s$. Let $\mu_e$ and $\mu_o$ be the largest eigenvalues in the even and odd sectors, and let $\nu_e$ be the second-largest even eigenvalue. The ground-state uniqueness result of Ref. [1] implies that the top eigenvalue of $K_s$ is $\mu_e$. Reversing the order of the spectrum when passing from $K_s$ to $H_s = -K_s$ gives

$$\Delta(T_q, s) = \mu_e - \max\{\mu_o, \nu_e\}. \tag{7}$$

It is therefore enough, and necessary for our method, to prove positivity of both margins

$$M_o(t) = \mu_e - \mu_o - (1-t), \qquad M_e(t) = \mu_e - \nu_e - (1-t). \tag{8}$$

The endpoint behavior illustrates why the strict statement uses the open interval. At $s = 0$, $K_0$ is diagonal and counts zero–zero edges; every finite tree has $\Delta = 1$. At $s = 1$ the target bound vanishes. Neither endpoint is included in Theorem 1.

## 3 Spin and staggered-charge reduction

Permutations of the $q$ ordinary leaves commute with $K_s$. Their Hilbert space decomposes into spin

$$j = \frac{q}{2} - r, \qquad 0 \le r \le \left\lfloor \frac{q}{2} \right\rfloor, \qquad d_{q,r} = \binom{q}{r} - \binom{q}{r-1}, \tag{9}$$

where $d_{q,r}$ is the multiplicity and $\binom{q}{-1} = 0$. In one copy of a spin sector, use the orthonormal basis $|c, a, b; k, r\rangle$, where $c, a, b \in \{0, 1\}$ are the bits at $C, A, B$, and $r \le k \le q - r$ is the physical Hamming weight of the ordinary leaves. The diagonal contribution from the $q$ ordinary edges is

$$q - k \quad (c = 0), \qquad tk \quad (c = 1), \tag{10}$$

and their nonzero off-diagonal matrix element is

$$|0, a, b; k, r\rangle \longleftrightarrow |1, a, b; k + 1, r\rangle, \qquad s\sqrt{(q - r - k)(k - r + 1)}. \tag{11}$$

The edges $CA$ and $AB$ add the same two-state action as Eq. (6) on the corresponding core bits.

There is a further reduction that uses bipartiteness. Assign positive sign to $L_i, A$, negative sign to $C, B$, and define

$$Q = k + a - c - b. \tag{12}$$

Every off-diagonal transition flips $00 \leftrightarrow 11$ across the bipartition, so $Q$ is conserved. Moreover, $(k+a+c+b) - Q = 2(c+b)$, hence parity equals $Q$ mod 2. For fixed $(r, Q)$, Eq. (12) determines $k = Q - a + c + b$ from three core bits. Each real symmetric block $B_{q,r,Q}(t)$ therefore has dimension at most eight, independently of $q$.

The complete parity characteristic polynomial factors as

$$\chi_p(x, t) = \prod_{r=0}^{\lfloor q/2 \rfloor} \prod_{Q \equiv p \pmod 2} \det[xI - B_{q,r,Q}(t)]^{d_{q,r}}. \tag{13}$$

The dimension identity

$$\sum_r d_{q,r} \sum_Q \dim B_{q,r,Q} = 2^{q+3} \tag{14}$$

is checked symbolically for every certified $q$, so no permutation or parity channel is omitted. Since each parity matrix is entrywise nonnegative, a top eigenvector can be chosen nonnegative and averaged over leaf permutations without vanishing. Thus the top eigenvalue in either parity lies in $r = 0$. At $s = 0$, the unique top state is the all-zero state in $(r, Q) = (0, 0)$. Ground-state simplicity for $0 \le s < 1$ prevents this root from changing invariant sectors. Consequently the physical ground root is always the largest root of the $(0, 0)$ block. For $q \ge 2$, after an identically zero factor is removed, its characteristic polynomial has degree six.

## 4 Analytic proof for $T_1 = P_4$

For the four-vertex path, exact construction of the parity blocks gives

$$\chi_e(x) = x^3(x - 1 - t)Q_t(x), \tag{15}$$

where

$$\begin{aligned} Q_t(x) = {} & x^4 - 5(1 + t)x^3 + (7 + 18t + 7t^2)x^2 \\ & - 3(1 + t)(1 + 4t + t^2)x + 2t(1 + 4t + t^2), \end{aligned} \tag{16}$$

and

$$\chi_o(x) = x^2(x-1-t)^2\left[x^2 - 2(1+t)x + 2t\right]^2. \tag{17}$$

The largest odd eigenvalue is

$$b(t) = 1 + t + r, \qquad r = \sqrt{1+t^2}. \tag{18}$$

All roots of $Q_t$ are real because it is a factor of the characteristic polynomial of a real symmetric invariant block. At $t = 0$, $Q_0 = x(x-1)^2(x-3)$, so exactly one root of $Q_0$ lies above $b(0) = 2$. Reduction modulo $r^2 = 1 + t^2$ yields the exact identity

$$Q_t(b(t)) = -(1-t)^2(1+t)b(t) < 0 \qquad (0 < t < 1). \tag{19}$$

No root can cross the continuous threshold $b(t)$ without making the left side zero. Hence exactly one even root remains above $b(t)$, and the first excitation is odd throughout the interval.

At the target-shifted odd root, the second exact identity is

$$Q_t\big(b(t) + 1 - t\big) = Q_t(2 + r) = -8t(1-t)\big[(1-t)r + t^2 - t + 1\big] < 0. \tag{20}$$

Since $Q_t$ is monic, its largest root is strictly larger than $b(t) + 1 - t$. That root is $\mu_e$, so

$$\Delta(P_4, s) = \mu_e - b(t) > 1 - t = 1 - s^2. \tag{21}$$

This proves Theorem 1 for $q = 1$.

## 5 Exact crossing certificates for $2 \le q \le 6$

This section describes the finite, independently checkable certificate used for the remaining cases. Let $P_q(x,t)$ be the nonzero degree-six factor containing the ground root, and let $F_{q,r,Q}(x,t)$ run over all nonzero charge–spin factors in (13). The ground factor itself is included when comparing its lower roots. All these polynomials are computed over $\mathbb{Q}[t,x]$.

If either margin in (8) vanished at an interior parameter, then for some physical competitor $\lambda$,

$$P_q(\mu_e, t) = 0, \qquad F_{q,r,Q}\big(\mu_e - (1-t), t\big) = 0. \tag{22}$$

Thus $t$ must be a zero of the exact resultant

$$R_{q,r,Q}(t) = \mathrm{Res}_x\left(P_q(x,t), F_{q,r,Q}(x-(1-t),t)\right). \tag{23}$$

For every listed block, this resultant is nonzero. Powers of $t$ and $t-1$ are removed because the theorem concerns the open interval, and the remaining univariate polynomial is replaced by its square-free part. Exact real-root isolation then returns every root in $(0,1)$. Across $q = 2, \ldots, 6$, retaining duplicates between different resultants, there are 171 candidate intervals.

At the rational midpoint $t_0$ of each isolating interval $[a,b]$, all real roots of all physical block polynomials are isolated over $\mathbb{Q}$. Taking a rational lower endpoint for the ground root and rational upper endpoints for every competing root gives an exact lower bound $m_0$ on

$$g_q(t_0) = \mu_e(t_0) - \max\{\mu_o(t_0), \nu_e(t_0)\} - (1 - t_0). \tag{24}$$

The pointwise bound must also control every parameter inside the isolating interval. Differentiating one edge term in the $(|00\rangle, |11\rangle)$ subspace gives

$$\left\| \frac{d}{ds} |\phi_s\rangle\langle\phi_s| \right\| = s + \sqrt{1+s^2} < \frac{5}{2}. \tag{25}$$

With $q+2$ edges, Weyl's inequality therefore makes every ordered eigenvalue $L_q$-Lipschitz in $s$, where

$$L_q = \frac{5}{2}(q+2). \tag{26}$$

The maximum of the competitors has the same Lipschitz constant. If $t_0 = (a+b)/2$, then

$$|\sqrt{t} - \sqrt{t_0}| = \frac{|t-t_0|}{\sqrt{t}+\sqrt{t_0}} \le \frac{b-a}{4a}. \tag{27}$$

Combining the variation of two eigenvalues with that of $1-t$ gives the rational enclosure

$$|g_q(t) - g_q(t_0)| \le \frac{b-a}{2}\left(\frac{L_q}{a}+1\right). \tag{28}$$

For every one of the 171 candidate intervals, the exact lower bound $m_0$ is strictly larger than the right side of Eq. (28). Table 1 summarizes the inventory. The displayed decimals are readability aids; every comparison is stored and checked as a rational inequality.

Table 1: Exact certificate summary. "Blocks" counts nonzero charge–spin factors. Candidate intervals retain duplicates across resultants. The reference margin is at $t = 1/4$.

| $q$ | $n$ | blocks | intervals | reference margin | minimum slack |
|---|---|---|---|---|---|
| 2 | 5 | 8 | 18 | 0.1072347684 | 0.0196245177 |
| 3 | 6 | 10 | 21 | 0.0668105392 | 0.0079834888 |
| 4 | 7 | 16 | 33 | 0.0486643413 | 0.0041072985 |
| 5 | 8 | 19 | 41 | 0.0383207710 | 0.0024037224 |
| 6 | 9 | 26 | 58 | 0.0316288710 | 0.0010475231 |

Exact root isolation also gives $g_q(1/4) > 0$ for every $q$ in the table. If $g_q$ vanished anywhere in $(0,1)$, Eq. (23) would place that point in a certified neighborhood on which $g_q > 0$, a contradiction. Continuity and the positive reference sign therefore prove $g_q(t) > 0$ throughout $(0,1)$. Identically zero eigenvalue factors cannot be competitors: the all-zero trial state has Rayleigh quotient $q+2$, so $\mu_e - 0 > 1 - t$. This completes the proof of Theorem 1.

### 5.1 Independent representation check

The exact generator uses fixed-size collective-weight matrices and checks Eq. (14). A separate checker constructs both parity matrices directly in the full computational basis from the local action $00 \leftrightarrow 11$. It verifies every saved rational neighborhood inequality and compares the full-basis spectrum at the exact reference point and the least-slack midpoint for each $q$. These floating-point diagonalizations corroborate the representation but are not used to prove the all-parameter result. The proof rests on the exact polynomials, resultants, rational isolators and enclosure (28). The symbolic calculations use SymPy [5]; the software environment and commands are specified in the accompanying reproducibility notes.

## 6 Physical interpretation and limitations

The star value $1-s^2$ is the known tight benchmark for the conjectured graph-independent lower bound [1]. Theorem 1 shows that subdividing one star edge does not create another equality case for the first six members of this family: on four through nine qubits the finite-volume gap is pointwise strictly larger for every $0 < s < 1$. In an adiabatic interpolation that stays

inside this finite model, these graphs therefore have a larger instantaneous gap than the star benchmark at the same $s$.

Several qualifications are essential. First, the strict excess is not uniform on the closed parameter interval; it can vanish as an endpoint is approached. Second, the cutoff $q \leq 6$ is arbitrary and the certificate does not establish the whole subdivided-star family. Third, no claim is made for an arbitrary tree, a graph with cycles, a weighted interaction, a normalized projector, a phase-deformed edge term, or the thermodynamic limit. Finally, a finite-tree result alone does not supply a new algorithm for arbitrary bipartite Quantum Max-Cut. Broader algorithmic consequences now follow from the general result of Ref. [3], by a route different from this finite-family certificate.

The block reduction suggests a sharper next step. Nonsymmetric leaf-spin sectors reduce to smaller symmetric problems with a positive but nonscalar field at $C$. A branch-selective Schur-complement or interlacing argument could compare the $(r, Q) = (0, 0)$ Perron root directly with the leading $(0, 1)$, $(0, 2)$, and recursive $r > 0$ blocks. Such an argument would avoid the irrelevant crossings that cause the resultant inventory to grow, and could either prove the result for every $q$ or isolate the first exact failure. That extension remains open.

# 7 Conclusion

We converted sampled evidence for six small non-star trees into an exact continuous-parameter statement. The combination of a fixed-size spin–charge decomposition, exact crossing resultants, rational root isolation and Weyl enclosures proves $\Delta(T_q, s) > 1 - s^2$ for $q = 1, \ldots, 6$ and all real $0 < s < 1$. The method controls all excitation sectors and distinguishes its exact certificate from a finite grid scan. The universal tree bound and its complete equality classification remain unresolved.

# Data and code availability

The Python generator, exact JSON certificate, independent full-basis checker, environment specification and regeneration instructions are included as supplementary files with this draft. The certificate uses open rational formats and does not require proprietary data. A permanent public repository and identifier remain to be selected by the author before submission.

# Generative-AI disclosure

OpenAI Codex using the `gpt-5.6-sol` model assisted with problem development, literature-search support, symbolic derivations, code generation and execution, same-machine checking, and manuscript drafting. This local draft and its calculations require critical review and final approval by the named author before submission. The AI system is not an author.

# Declaration of interests

The declaration of interests is pending confirmation by the named author.

# References

[1] Benjamin Wong, Sergey Bravyi, David Gosset, and Yinchen Liu. Lee–Yang tensors and hamiltonian complexity, 2026. arXiv:2602.03605v1.

[2] Masuo Suzuki and Michael E. Fisher. Zeros of the partition function for the Heisenberg, ferroelectric, and general Ising models. *Journal of Mathematical Physics*, 12(2):235–246, 1971.

[3] Sergey Bravyi, David Gosset, Yinchen Liu, and Benjamin Wong. Efficient quantum algorithm for Heisenberg spin systems, 2026. arXiv:2607.14401v1.

[4] Chaithanya Rayudu and Jun Takahashi. Spectral gap of Lee–Yang hamiltonians, 2026. arXiv:2607.10765v1.

[5] Aaron Meurer, Christopher P. Smith, Mateusz Paprocki, Ondřej Čertík, Sergey B. Kirpichev, Matthew Rocklin, AMiT Kumar, Sergiu Ivanov, Jason K. Moore, Sartaj Singh, Thilina Rathnayake, Sean Vig, Brian E. Granger, Richard P. Muller, Francesco Bonazzi, Harsh Gupta, Shivam Vats, Fredrik Johansson, Fabian Pedregosa, Matthew J. Curry, Andy R. Terrel, Štěpán Roučka, Ashutosh Saboo, Isuru Fernando, Sumith Kulal, Robert Cimrman, and Anthony Scopatz. SymPy: symbolic computing in Python. *PeerJ Computer Science*, 3:e103, 2017.